Huaiyuan Jing

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Education

University of Illinois Urbana-Champaign — Urbana-Champaign, IL
Master of Computer Science — *Aug 2026 — Present*

University of Wisconsin-Madison — Madison, WI
B.S. Computer Science; B.S. Mathematics — *Sep 2022 — May 2026*

- GPA: **3.8/4.0** | Authorized to work in the United States

Experience & Research

Founding CTO / Technical Co-Founder — Sep 2026 — Present
Intgral — *Remote*

- Lead infrastructure and engineering for an AI-assisted e-commerce platform built on Medusa ERP, supporting agent access to product and listing workflows through a Model Context Protocol (MCP) gateway.
- Own the multi-merchant infrastructure roadmap, including identity and access control, per-tenant instance and database isolation, background task execution, and failure recovery.
- Deployed self-hosted GitHub Actions runners and a webhook-driven autoscaler on Railway; configured runner registration and cleanup to support compute-intensive CI workloads.
- Established CI checks for linting, builds, tests, and security scanning; deployed a webhook-based PR review agent using the GitHub API to support code review in an AI-assisted development workflow.

First Author — Summer 2026
Quantum Computing Algorithm Optimization Research — *Madison, WI*

- Developed a polar-decomposition-based construction for **single-ancilla block encoding**, reducing auxiliary-qubit requirements to one ancilla.
- Analyzed coherent ancilla reuse and approximation error for a **$(1, 1, \varepsilon)$-block encoding**, allocating an $\frac{\varepsilon}{2}$ error budget to each of two branches.
- Derived query-complexity and error bounds, including a polynomial-degree bound of $O\left(\frac{1}{\delta_{\text{abs}}} \log\left(\frac{1}{\eta_{\text{abs}}}\right)\right)$ for absolute-value approximation.

Research Assistant — Feb 2024 — Sep 2024
Madison Experimental Mathematics (MXM) Lab — *Madison, WI*

- Built a concurrent Rust toolkit for graph hashing and combinatorial game analysis using Grundy numbers; evaluated **50 data points** in workloads with $O(2^m)$ time complexity.
- Used Rust's ownership model and Send/Sync traits to structure memory-safe parallel computation; optimized graph traversal and synchronization with lock-free data structures.

Projects

Independent Developer, High-Frequency Limit Order Book — codeberg.org/HYJING/hft_orderbook

- Implemented a fixed-point limit order book and matching engine in **C++20**; benchmarked throughput of **55 million orders per second** with Google Benchmark on an 8-core CPU system.
- Combined ordered price-level maps with intrusive doubly linked FIFO queues, enabling constant-time queue insertion and removal once the price level and order are located.
- Implemented a vector-backed object pool to reuse order storage and reduce heap-allocation overhead in the matching path.

Independent Developer, Linux Security Boot Module — github.com/melody0123/dracut-cryptsetup-duress

- Maintained a Dracut initramfs module for LUKS-encrypted systems, working with early-boot cryptsetup integration and Linux boot-time scripting.

Honors & Awards

Gold Medal, ICPC North Central North America Regional — 2024--2025

Silver Medal, ICPC North Central North America Regional — 2025--2026